# Inorganic CUME on Chalcogenides

Oct-2-2015

**1. Which group(s) would have all three compounds called chalcogenides? (10 points)**

a) ZnS, ZnSe, ZnO

b) ZnTe, CdS, CuSe

c) InP, GaAs, SiC

d) CuO, CuS, CuSe

**2. Synthesis, characterization, property, and application: (10 points each)**

a) Propose a facile way for making one of the chalcogenides in the above.

b) Briefly describe a method for confirming the chalcogenide's chemical composition.

c) What method(s) could you use for determining the chalcogenide's crystal structure?

d) Briefly describe a method for testing the chalcogenide's electrical conductivity.

e) What method(s) would you use to characterize the chalcogenide's particle size and shape?

f) Briefly discuss which of the abovementioned chalcogenides can be toxic.

g) Which chalcogenide(s) in the above could be air-sensitive?

h) List all in the above that can be water-soluble.